

FUNCTIONAL SPECIFICATION

PDF TO HL7 CONVERTER V0.1.0

Document Version: 1.0 **Date:** 23 February 2026 **Prepared by:** SMEC AI **Client:** BJC Health

1. OVERVIEW

The PDF to HL7 Converter is a web application that converts patient PDF documents into Australian HL7 v2.4 messages compatible with Genie clinical software. The system extracts patient data from uploaded PDFs using pattern matching, generates standards-compliant HL7 messages with the original PDF embedded as Base64, and provides a download for import into practice management systems.

1.1 Key Capabilities

- Automatic document type detection (consent forms, specialist referrals, GP referrals)
- Patient data extraction via regex pattern matching (name, DOB, sex, Medicare, address, phone)
- HL7 v2.4 ORU^R01 message generation per Australian ADRM specification
- PDF embedding as Base64 in OBX segment for Genie import
- Password-protected web interface with drag-and-drop upload
- Configurable auto-filing and doctor routing options

1.2 Technology Stack

Component	Technology
Framework	Next.js 14 (App Router)
Runtime	Bun
Language	TypeScript 5
PDF Parsing	pdf-parse (Poppler-based)
Styling	Tailwind CSS 3 + Custom CSS
Deployment	AWS Amplify (WEB_COMPUTE / SSR)
Testing	Bun test runner (244 tests)

2. SYSTEM ARCHITECTURE

2.1 Data Flow



2.2 Component Overview

Module	Purpose
<code>app/page.tsx</code>	Main converter UI with drag-drop upload
<code>app/login/page.tsx</code>	Password authentication page
<code>app/api/convert/route.ts</code>	PDF conversion API endpoint
<code>app/api/auth/route.ts</code>	Authentication API endpoint
<code>lib/pdf-parser.ts</code>	PDF text extraction and patient data parsing
<code>lib/hl7-builder.ts</code>	HL7 v2.4 message generation
<code>middleware.ts</code>	Route protection and session validation

3. AUTHENTICATION

3.1 Authentication Flow

The application uses simple password-based authentication with HTTP-only cookies.

Login Flow:

1. User navigates to any protected route
2. Middleware checks for `app_authenticated` cookie
3. If absent, redirects to `/login`
4. User enters password on login page
5. `POST /api/auth` validates against `APP_PASSWORD` environment variable
6. On success, sets `app_authenticated` cookie (HTTP-only, 7-day expiry)
7. User redirected to home page

Logout Flow:

1. `DELETE /api/auth` clears the authentication cookie
2. User redirected to login page

3.2 Cookie Configuration

Property	Value
Name	app_authenticated
Value	"true"
httpOnly	true
secure	true (production) / false (development)
sameSite	"lax"
maxAge	604,800 seconds (7 days)
path	/

3.3 Protected Routes

All routes are protected except:

- /login - Login page
- /api/auth - Authentication endpoint
- Static assets (_next/static , _next/image , images, favicon)

4. API ENDPOINTS

4.1 POST /api/convert - PDF Conversion

Converts a PDF document to an HL7 v2.4 message with the original PDF embedded.

Request: Multipart form data (multipart/form-data)

Parameter	Type	Required	Default	Description
pdf	File	Yes	-	PDF file to convert (max 10MB)
detectOnly	string	No	"false"	If "true", returns detected document type only
documentType	string	No	"auto"	"auto", "consent_form", "referral_letter", "gp_referral"
autoFile	string	No	"true"	"true" = auto-file (Final), "false" = queue for review (Preliminary)
orderingProvider	string	No	-	Medicare Provider Number (e.g., "1234567A") for doctor routing

Success Response (200):

```
{
  "success": true,
  "filename": "LASTNAME_FIRSTNAME_20260223143045.hl7",
  "hl7Content": "[HL7 message with CR segment terminators]",
  "extractedData": {
    "firstName": "John",
    "lastName": "Smith",
    "dob": "23/02/1980",
    "sex": "Male",
    "medicareNo": "1234567890-1"
  },
  "warnings": []
}
```

Error Responses:

Status	Condition	Message
400	No file uploaded	"No PDF file provided"
400	Wrong file type	"File must be a PDF"
400	File too large	"File size exceeds 10MB limit"
500	Processing error	"Conversion failed" or specific error

4.2 GET /api/convert - Health Check

Returns service status.

Response (200):

```
{
  "status": "ok",
  "service": "PDF to HL7 Converter",
  "version": "1.0.0"
}
```

4.3 POST /api/auth - Login

Validates password and creates session.

Request Body: { "password": "string" }

Success Response (200): { "success": true } (sets authentication cookie)

Error Responses:

Status	Condition	Message
401	Wrong password	"Invalid password"
500	No APP_PASSWORD set	"Server configuration error"

4.4 DELETE /api/auth - Logout

Clears session cookie.

Response (200): { "success": true } (clears authentication cookie)

5. DOCUMENT TYPE DETECTION

5.1 Supported Document Types

Type	Identifier	Description	Example Source
Consent Form	consent_form	Patient information and consent forms	BJC Health intake forms
Specialist Referral	referral_letter	Specialist referral letters	NeuroSpine, specialist clinics
GP Referral	gp_referral	General practitioner referral letters	Best Practice exports

5.2 Detection Logic

Auto-detection examines extracted PDF text using this decision tree:

1. Check for referral indicators:
 - "Dear Dr/Professor" OR "Dear [Name]," present?
 - "RE:" or "re." present?
2. If BOTH present --> Referral type
 - Check for GP format: "re. Mr/Mrs/Miss/Ms [Name]"
 - Check for Medicare number in text
 - If GP format OR Medicare present --> gp_referral
 - Otherwise --> referral_letter
3. If NOT both present --> consent_form

5.3 Detection Patterns

Referral indicators:

- **Dear Dr/Professor** : `/Dear\s+(?:Dr|Professor)/i`
- **Dear [Name]** : `/Dear\s+[A-Z][a-z]+,/m`
- **RE/re line** : `/\b(?:RE|re)[:\.\.]\s+/i`

GP Referral distinguishers:

- **Title + name format**: `/\bre\.\s+(?:Mr|Mrs|Miss|Ms|Dr)\s+[A-Za-z]+\s+[A-Za-z][A-Za-z'-]*\s*/i`
 - **Medicare presence**: `/Medicare\s*No[:\s]+\d{10,11}/i`
-

6. PATIENT DATA EXTRACTION

6.1 Extracted Fields

Field	Type	Required	Default	Description
firstName	string	Yes	"UNKNOWN"	Patient first name
lastName	string	Yes	"PATIENT"	Patient last name
dob	string	Yes	"19000101"	Date of birth (YYYYMMDD)
sex	string	Yes	"U"	"M", "F", or "U"
address	string	No	-	Street address
suburb	string	No	-	Suburb/city
state	string	No	-	3-letter state code
postcode	string	No	-	4-digit postcode
phone	string	No	-	Phone number
medicareNo	string	No	-	10-11 digit Medicare number
medicareRef	string	No	-	Single reference digit

6.2 Consent Form Extraction

Parses BJC Health consent forms with labelled fields:

Field	Pattern	Example Match
Title	<code>/^\s*(Mr Mrs Miss Ms)\s*\$/m</code>	Mr
First Name	<code>/First Name\s**\s*\n?\s*([A-Za-z]+)/i</code>	John
Last Name	<code>/Last Name\s**\s*\n?\s*([A-Za-z]+)/i</code>	Smith
DOB	<code>/Date of Birth\s**\s*\n?\s*(\d{1,2}\s*\d{1,2}\s*\d{4})/i</code>	23/02/1980
Mobile	<code>/Mobile Phone\s**\s*\n?\s*(\d\s*){10,14}/i</code>	0412 345 678
Medicare No	<code>/Medicare Card No\.\s**\s*\n?\s*(\d{10,11})/i</code>	2673291844
Medicare Ref	<code>/Medicare Ref\s*(?:Number)?\s**\s*\n?\s*(\d)/i</code>	1
Address	<code>/Address\s**\s*\n?\s*(.+?)(?=\n*Postcode \n*City)/is</code>	42 George St
Postcode	<code>/Postcode\s**\s*\n?\s*(\d{4})/i</code>	2000
Suburb	<code>/City\s*\s*\n?\s*\s*\n?\s*(\s*[A-Za-z\s]+?)(?=\n State)/i</code>	Sydney

6.3 Specialist Referral Extraction

Parses specialist referral letters (e.g., NeuroSpine format):

Primary pattern - RE: line with DOB:

```
/RE:\s*(?:([A-Za-z][A-Za-z'-]*)\s+([A-Z][A-Z'-]+)|([A-Z][A-Z'-]+),\s*([A-Za-z][A-Za-z'-]*)\s*[-]\s*DOB:\s*(\d{1,2}\s*\d{1,2}\s*\d{4})/i
```

Matches both formats:

- RE: Scott LAWLER - DOB: 15/03/1985
- RE: LAWLER, Scott - DOB: 15/03/1985

Secondary patterns:

Field	Pattern
Phone	<code>/(\?:Mobile Ph Tel Phone)[:\s]+(\d[\d]{9,14})/i</code>
Address	<code>/^\s*(\d+[\^,\n]+),\s*([A-Z][A-Za-z\s]+),\s*([A-Z]{2,3}),?\s*(\d{4})\s*\$/m</code>
Provider No	<code>/Provider\s*No[:\.]?\s*(\d{6}[A-Z]{2})/i</code>
Letter Date	<code>/(\d{1,2}\s+(\?:January February ...)\s+\d{4})/i</code>

6.4 GP Referral Extraction

Parses Best Practice GP referral letters:

Primary pattern - re. line with title:

```
/\bre\.\s+(Mr|Mrs|Miss|Ms|Dr)\s+([A-Za-z]+)\s+([A-Za-z][A-Za-z'-]*)/i
```

Example: `re. Mr Tim Ball`

Additional patterns:

Field	Pattern
DOB	<code>/^bDOB:\s*(\d{1,2}\s*\d{1,2}\s*\d{4})/i</code>
Medicare	<code>/Medicare\s*No[:\s]+(\d{10,11})/i</code>
Mobile	<code>/Mobile:\s*(\d[\d]{10,14})/i</code>
Provider No	<code>/^\s*(\d{6}[A-Z]{2})\s*\$/m</code>

Address extraction is scoped to text between the `re.` line and `Dear` line to avoid matching clinic letterhead addresses.

6.5 Sex Determination

Priority order:

- Title mapping:** Mr=M, Mrs/Miss/Ms=F, Dr/Mx=U
- Pronoun inference:** Counts `he/him/his` vs `she/her/hers` in document text. Requires 2+ pronouns in dominant category for reliable inference.
- Default:** `"U"` (Unknown)

6.6 State Inference from Postcode

Maps first digit of 4-digit Australian postcode to state:

First Digit	State
2	NSW
3	VIC
4	QLD
5	SA
6	WA
7	TAS
0	NT

Default: VIC (if postcode invalid or unavailable)

7. HL7 MESSAGE GENERATION

7.1 Message Format

Property	Value
Standard	HL7 v2.4
Message Type	ORU^R01 (Observation Result, Unsolicited)
Specification	Australian ADRM
Segment Terminator	<code>\r</code> (CR only, no LF)
Encoding	ISO 8859/1 (Latin-1)

HL7 Encoding Characters:

Character	Purpose	Escape
<code> </code>	Field separator	<code>\F\</code>
<code>^</code>	Component separator	<code>\S\</code>
<code>~</code>	Repetition separator	<code>\R\</code>
<code>\</code>	Escape character	<code>\E\</code>
<code>&</code>	Subcomponent separator	<code>\T\</code>

7.2 MSH - Message Header

MSH|^~\&|MEDHOST|BJCHEALTH|GENIE|CLINIC|{timestamp}||ORU^R01|{messageId}|P|2.4||AL|INE|AUS|8859

Field	Value	Description
MSH-3	MEDIHOST	Sending application
MSH-4	BJCHEALTH	Sending facility
MSH-5	GENIE	Receiving application
MSH-6	CLINIC	Receiving facility
MSH-7	YYYYMMDDHHMMSS	Message timestamp
MSH-9	ORU^R01	Message type
MSH-10	MSG{timestamp}{4 random chars}	Unique message ID
MSH-11	P	Processing ID (Production)
MSH-12	2.4	HL7 version
MSH-15	AL	Accept acknowledgement (logging level)
MSH-16	NE	No application acknowledgement
MSH-17	AUS	Country code
MSH-18	8859/1	Character set

7.3 PID - Patient Identification

PID|1||{medicare}-{ref}^^Medicare^MC||{lastName}^{firstName}||{dob}|{sex}|||{address}^^{suburb}

Field	Description	Format
PID-3	Medicare identifier	{number}-{ref}^^Medicare^MC
PID-5	Patient name	{lastName}^{firstName}
PID-7	Date of birth	YYYYMMDD
PID-8	Sex	M, F, or U
PID-11	Address	{street}^^{suburb}^{state}^{postcode}
PID-13	Phone	Digits only

7.4 PV1 - Patient Visit

Minimal format (no provider): PV1|1|10

With ordering provider: PV1|1|10|||||{providerNo}^^AUSHICPR

Field	Description
PV1-2	Patient class: 0 (Outpatient)
PV1-9	Consulting doctor: Medicare Provider Number with AUSHICPR identifier

Only included when **orderingProvider** parameter is provided.

7.5 OBR - Observation Request

OBR|1||RPT{timestamp}^MEDIHOST|PDF^{title}^L||||{timestamp}|||||||{timestamp}||{status}

Field	Description
OBR-3	Filler order number: RPT{timestamp}^MEDIHOST
OBR-4	Universal service ID: PDF^{document title}^L
OBR-7	Observation date/time
OBR-22	Results change date/time
OBR-25	Result status: F (Final/auto-file) or P (Preliminary/queue)

7.6 OBX - Observation/Result (Embedded PDF)

OBX|1|ED|PDF^Display format in PDF^AUSPDI||^application^pdf^Base64^{base64data}|||||F

Field	Description
OBX-2	Value type: ED (Encapsulated Data)
OBX-3	Observation ID: AUSPDI coding system
OBX-5	PDF as Base64 in ED format: ^application^pdf^Base64^{data}
OBX-11	Observation result status: F (Final)

7.7 HL7 Options

Option	Default	Description
sendingApplication	"MEDIHOST"	MSH-3 value
sendingFacility	"BJCHEALTH"	MSH-4 value
receivingApplication	"GENIE"	MSH-5 value
receivingFacility	"CLINIC"	MSH-6 value
documentTitle	"Patient Consent Form"	OBR-4 display name
resultStatus	"F" or "P"	Auto-file vs queue for review
orderingProvider	-	Medicare Provider Number for doctor routing

7.8 Filename Generation

Format: {LASTNAME}_{FIRSTNAME}_{YYYYMMDDHHMMSS}.hl7

Non-alphanumeric characters in names are replaced with underscores.

8. USER INTERFACE

8.1 Login Page (/login)

A single password input form with BJC Health and SMEC AI branding.

Elements:

- Logo strip (BJC Health + SMEC AI logos)
- Password input field (placeholder: "Enter access password")
- Submit button (disabled while loading or empty)
- Error message display
- Contact admin footer text

Behaviour:

- Auto-redirects authenticated users to home page
- Displays error on invalid password
- Clears password field on failed attempt
- Shows loading spinner during authentication

8.2 Home Page (/) - Converter

The main converter interface with drag-and-drop upload and conversion options.

Upload Zone:

- Drag-and-drop area with visual feedback (border colour change on drag)
- File picker button ("Browse")
- Accepts .pdf files only
- Shows file name, size, and remove button when file selected

- 10MB file size limit (validated server-side)

Auto-Detection:

- Triggers on file selection or drop
- Sends `detectOnly=true` request to API
- Shows "detecting..." badge during request
- Shows "auto-detected" badge on completion

Conversion Options (shown after file selection):

Option	Type	Default	Description
Document Type	Select	Auto-detect	Choose: Auto, Consent Form, Specialist Referral, GP Referral
Auto-file	Checkbox	Checked	When checked, document auto-files in Genie. When unchecked, queues for review
Send to Doctor	Checkbox	Unchecked	Enables provider number input
Provider Number	Text input	Empty	Medicare Provider Number for routing (shown only if Send to Doctor checked)

Convert Button:

- Full-width primary style
- Shows spinner and "Converting..." text during processing
- Disabled while converting

Success Display:

- Green success box with checkmark
- Patient data grid: Name, DOB, Sex, Medicare number
- "Download HL7 File" button (green)
- "Convert Another" button (secondary)

Error Display:

- Red error box with icon
- Error message from API

8.3 Visual Design

Colour Palette:

Colour	Hex	Usage
BJC Navy	#002e50	Button hover, headings
BJC Blue	#2874cc	Primary buttons, focus states, links
BJC Teal	#01bfa5	Accent elements
BJC Orange	#fe7b29	Warning states
BJC Green	#5bb841	Success indicators
Success	#16a34a	Success messages, download button
Error	#dc2626	Error messages

Typography:

- Primary: Plus Jakarta Sans (weights 300–800)
- Monospace: JetBrains Mono (weights 400–500)

Components: Cards with 16px border radius and subtle shadows, custom checkboxes, animated fade-in-up transitions with staggered delays.

9. DEPLOYMENT

9.1 AWS Amplify Configuration

Setting	Value
Platform	WEB_COMPUTE (required for SSR)
Build output	.next directory
Build format	frontend: (not applications:)
Runtime	Node.js with standalone output

9.2 Environment Variables

Variable	Location	Purpose
APP_PASSWORD	Amplify Console	Application password

Build phase creates .env.production from Amplify env vars to ensure they’re available at Lambda runtime.

9.3 Build Commands

```
bun dev           # Development server (localhost:3000)
bun run build     # Production build
bun run lint      # ESLint check
bun test          # Run test suite (244 tests)
bun start         # Production server
```

10. CONSTRAINTS AND LIMITATIONS

10.1 Technical Constraints

Constraint	Detail
PDF type	Machine-readable text PDFs only (no OCR)
File size	Maximum 10MB per upload
Character set	ISO 8859/1 (Latin-1) - non-ASCII characters escaped
Concurrency	Single-threaded Node.js, no job queue
Scale	Suitable for fewer than 100 PDFs per day

10.2 Authentication Constraints

Constraint	Detail
Auth model	Single shared password (no multi-user support)
Session	Fixed 7-day expiry, no refresh mechanism
Password storage	Plaintext comparison (not hashed)
Audit	No access logging

10.3 Data Extraction Constraints

Constraint	Detail
Extraction method	Regex pattern matching (not AI/ML)
Document formats	Three supported formats only
Missing fields	Defaults used when extraction fails (warnings generated)
Address parsing	Scoped to patient block to avoid letterhead addresses
Medicare ref	Defaults to "1" if not found

10.4 HL7 Constraints

Constraint	Detail
Standard	HL7 v2.4 only
Message type	ORU^R01 only
Target system	Genie clinical software
Timestamp	Server local time (no timezone conversion)

11. TESTING

11.1 Test Coverage

Test File	Count	Coverage
lib/pdf-parser.test.ts	100+	All regex patterns, edge cases, special characters
lib/hl7-builder.test.ts	100+	All segments, encoding, escaping
app/api/convert/route.test.ts	40+	API validation, integration flows
Total	244	

11.2 Test Infrastructure

Script	Purpose
scripts/generate-test-pdfs.ts	Creates 20 test PDFs with realistic content
scripts/diagnose-pdfs.ts	Runs extraction across all test PDFs
scripts/dump-text.ts [file]	Dumps raw PDF text for regex debugging

11.3 Test PDF Variants

Generated test PDFs include: standard forms, skewed layouts, grainy/fax-quality, logo overlays, multicultural names (O'Brien, hyphenated surnames), and nested directory structures.

APPENDIX A: FILE STRUCTURE

```
medihost-pdf-to-hl7/
+-- app/
|   +-- api/
|   |   +-- auth/route.ts           Authentication endpoint
|   |   +-- convert/route.ts        Conversion endpoint
|   |   +-- convert/route.test.ts   Integration tests
|   +-- login/page.tsx              Login page
|   +-- layout.tsx                  Root layout
|   +-- page.tsx                    Converter page
|   +-- globals.css                 Styling
+-- lib/
|   +-- pdf-parser.ts               PDF extraction logic
|   +-- pdf-parser.test.ts          Parser tests
|   +-- hl7-builder.ts              HL7 generation
|   +-- hl7-builder.test.ts         Builder tests
+-- scripts/
|   +-- generate-test-pdfs.ts        Test PDF generator
|   +-- diagnose-pdfs.ts             Extraction diagnostics
|   +-- dump-text.ts                 Raw text dumper
+-- middleware.ts                   Auth middleware
+-- next.config.mjs                 Next.js config
+-- amplify.yml                     AWS Amplify config
+-- package.json                    Dependencies
```

APPENDIX B: API QUICK REFERENCE

Health check

```
curl http://localhost:3000/api/convert
```

Convert PDF (auto-detect type)

```
curl -X POST -F "pdf=@file.pdf" http://localhost:3000/api/convert
```

Detect document type only

```
curl -X POST -F "pdf=@file.pdf" -F "detectOnly=true" \  
http://localhost:3000/api/convert
```

Convert with options

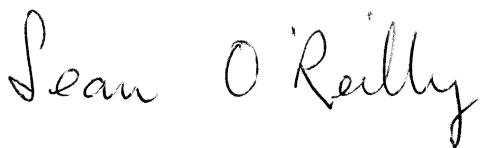
```
curl -X POST \  
-F "pdf=@file.pdf" \  
-F "documentType=gp_referral" \  
-F "autoFile=false" \  
-F "orderingProvider=1234567A" \  
http://localhost:3000/api/convert
```

Login

```
curl -X POST -H "Content-Type: application/json" \  
-d '{"password":"your_password"}' \  
http://localhost:3000/api/auth
```

Logout

```
curl -X DELETE http://localhost:3000/api/auth
```



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